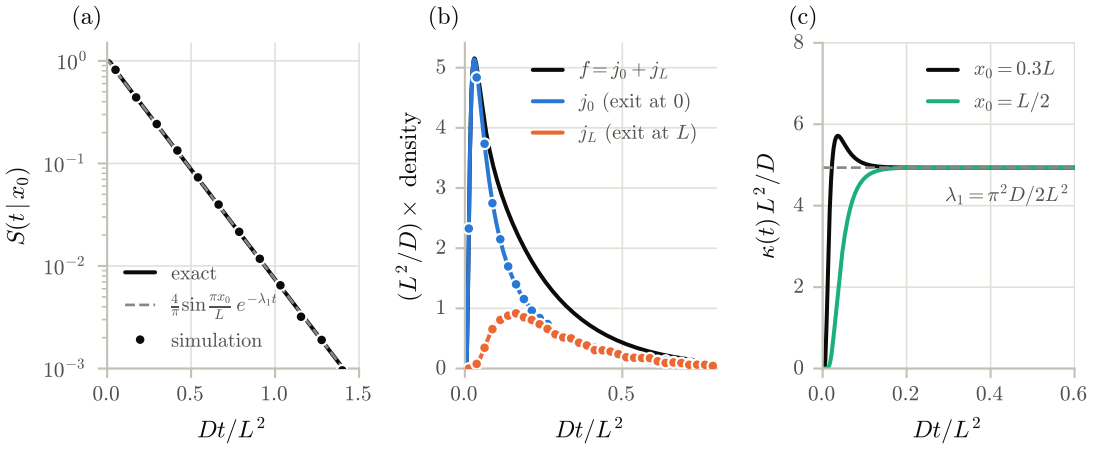
(a)
10^0
10^{-1}
10^{-2}
10^{-3}
$S(t\,|\,x_0)$
exact
$\frac{4}{\pi}\sin\frac{\pi x_0}{L}\,e^{-\lambda_1 t}$
simulation
0.0
0.5
1.0
1.5
Dt/L^2
(b)
5
4
3
2
1
0
$(L^2/D)\times$ density
$f = j_0 + j_L$
j_0 (exit at 0)
j_L (exit at L)
0.0
0.5
Dt/L^2
(c)
8
6
4
2
0
$\kappa(t)\,L^2/D$
$x_0 = 0.3L$
$x_0 = L/2$
$\lambda_1 = \pi^2 D/2L^2$
0.0
0.2
0.4
0.6
Dt/L^2